

Army OER forward vs backward — and why pool sizes look wrong

Last synced: 2026-09-20

Audience: Charles (re-entry after ~1 year away from the 520 pipeline code)

Standalone: Explains what `_fwd` and `_bwd` mean in the talent pipeline, which mode is used for own performance vs senior-rater pools in Run 1, why panel 6 pool sizes can exceed ~50, and a recommended fix — without opening the notebooks.

0. One-sentence version

Forward (`_fwd`) = “what OER history is **known by** this snapshot date.” Backward (`_bwd`) = “which OER rating window is active on this snapshot date.” Run 1 uses **forward for the officer’s own top-block ratio and for pool comparisons**, but uses **backward to pick *who* is in the senior-rater pool** — and then groups poolmates on **calendar snapshot day**, which is too coarse and inflates pool size.

1. The two clocks you are always juggling

Every officer row in the pipeline is tied to two different timelines:

Timeline	What it is	Example column
HRC snapshot date	A fixed day on the personnel system clock (<code>snpsht_dt</code>)	2018-03-31
OER rating period	Start and end of one evaluation (<code>eval_strt_dt</code> ... <code>eval_thru_dt</code>)	2017-10-01 ... 2018-09-30

An OER row in the raw file always has a **senior rater** (`snr_rater`) — that person is the SNR for **that evaluation period**, not necessarily “everyone this general ever rated.”
Cell 4 (`assign_oer_to_snapshots_fast`) stamps **both** forward and backward OER information onto every snapshot row. Cell 5–6 build **pools** from those stamps. Run 1 Cox/HERO mostly use the `_fwd` performance columns, but pool **membership** keys off `snr_rater_bwd` .

2. Forward vs backward — plain English

Think of one captain and one snapshot date **S**.

Forward (`_fwd`) — “last completed OER as of S”

Rule: Find the most recent OER whose **through date** (`eval_thru_dt`) is **on or before** S.

Intuition: “As of this snapshot, what is the **latest finished** evaluation the Army has on file?”

Simple example

Snapshot S	Last completed OER thru date	What <code>_fwd</code> sees
2018-03-31	2017-09-30 (done)	That OER’s rater, SNR, boxes
2018-03-31	(none yet)	Missing — officer not yet evaluated

Used for: cumulative top-block counts and `tb_ratio_fwd_snr` — the officer’s **own** performance track record **as known at S**.

Code: `merge_asof` with `direction='backward'` on `eval_thru_dt` → columns like `snr_rater_fwd` , `eval_thru_dt_fwd` , `tb_ratio_fwd_snr` .

Backward (`_bwd`) — “OER window that covers S”

Rule: Find an OER where **start ≤ S ≤ through** (snapshot falls **inside** the rating period). If S is before the period starts, fields are cleared.

Intuition: “On this snapshot date, which **active rating cycle** am I in?”

Simple example

Snapshot S	OER window	What <code>_bwd</code> sees
2018-03-31	2017-10-01 – 2018-09-30	SNR and rater for that cycle
2018-03-31	2018-10-01 – 2019-09-30 (not started yet)	Missing (S before <code>eval_strt_dt</code>)
2018-03-31	2016-10-01 – 2017-09-30 (already closed)	Not this window — forward might still use it

Used for: `snr_rater_bwd` , `eval_strt_dt_bwd` , `eval_thru_dt_bwd` — *who is rating me in the cycle that covers today?*

Code: `merge_asof` with `direction='forward'` on `eval_thru_dt` , then mask to the window → columns like `snr_rater_bwd` , `eval_thru_dt_bwd` .

Naming trap (read this twice)

In `pipeline_config.py` , the words “forward” and “backward” describe **two different merge strategies**. They do **not** mean “forward = future” in everyday speech.

Config label	Column suffix	Merge idea
TB_ASSIGNMENT_MODE = 'forward'	_fwd	Last completed eval through S
"backward assignment" in code comments	_bwd	Eval window covering S

You remembered correctly: **one mode for “my OER history” and another for “which rating pool am I in.”**

3. What Run 1 actually uses (today)

Defaults in `pipeline_config.py` + `pipeline_config_17_1.py` (Run 1):

TB_ASSIGNMENT_MODE_SELF	= 'forward'	→	tb_ratio_fwd_snr, z_tb_ratio_fwd_snr
TB_ASSIGNMENT_MODE_POOL	= 'forward'	→	pool_minus_mean_snr_fwd, z_pool_minus_mean_snr_fwd

So in Run 1:

Concept	Forward or backward?	Column(s)
Own performance (Â)	Forward	tb_ratio_fwd_snr → z_tb_ratio_fwd_snr
Pool comparison (LOO)	Forward (peer ratios)	pool_minus_mean_snr_fwd → z_pool_minus_mean_snr_fwd
Who is my senior rater for pooling?	Backward ID	snr_rater_bwd
Pool size on BDP panel 6	Forward suffix, backward key	pool_size_snr_fwd grouped by (snpsht_dt, snr_rater_bwd)

That last row is the subtle part you half-remembered:

- **Peer performance** in the pool mean is `tb_ratio_fwd_snr` (forward).
- **Pool roster key** is `snr_rater_bwd` (backward SNR person ID).
- **Pool roster also requires same** `snpsht_dt` — every officer in the cohort on that **calendar snapshot day** who shares that SNR ID.

Cell 5 calls this explicitly:

```
add_pool_means_and_sizes(  
    ...  
    rtr_col='rtr_rater_bwd',  
    snr_col='snr_rater_bwd',           # ← pool membership key (backward)  
    ratio_snr_fwd_col='tb_ratio_fwd_snr', # ← value averaged in pool (forward)  
    ...  
)
```

512 OER ingest still drops rows with no `snr_rater` — that part is fine. The pool-size issue is **not** “we forgot to drop null SNRs”; it is **how we define “same pool”** after that.

4. Tiny story — three officers, one general

General **G** is senior rater on several OERs. Captains **A, B, C** all have `snr_rater_bwd = G` on snapshot **2018-03-31**, but:

- **A’s** active window: FY17 cycle (thru 2017-09-30) — backward match might be FY18 cycle depending on dates
- **B** and **C** share the **same** FY18 window (thru 2018-09-30)

Under **current code**, if all three rows show `snr_rater_bwd = G` on **2018-03-31**, they land in **one pool**:

Pool key = (snpsht_dt = 2018-03-31, snr_rater_bwd = G)
Pool size = count of officers with valid tb_ratio_fwd_snr in that bucket

Army reality: G’s **FY18 rating pool** might be ~15 officers in one chain — not every captain in the entire dataset who ever shows G on a March snapshot.

That is why `pool_size_snr_fwd` can be 100+ while HRM’s `snr_rater_rates_this_grd` on the OER row is often ~12–25.

5. What your probe just showed (2026-09-20 AWS)

Measure	Median	95th pct	Max	Share > 50
Computed <code>pool_size_snr_fwd</code> (last snapshot)	23	79	264	19.1%
Official <code>snr_rater_rates_this_grd</code> (OER rows)	12	56	(<i>bad sentinel</i>)	6.4%
Alt count by <code>(snr_rater_bwd × eval_thru_dt_bwd)</code>	4	33	766	2.7%

How to read this

- 1. **Official field** — Your “pools aren’t bigger than ~50” instinct matches **p95 = 56** on real OER rows. Median **12** is plausible. Ignore absurd max (7e9) until filtered — data quality issue, not your code.
 - 2. **Computed pool** — **Too fat**. Median 23 and long tail above 50 = snapshot×SNR grouping is **not** the Army rating pool.
 - 3. **Alt grouping** — Adding `eval_thru_dt_bwd` to the pool key shrinks counts toward Army-like scale (median 4, p95 33). Still some monsters (766) — next refinement may need `eval_strt_dt_bwd` too, or UIC — but direction is clear.
- Conclusion:** Not a transcription bug; a **pool definition** bug. LOO and Cox pool context built on the fat pools are **methodologically soft** until re-keyed.

6. What each pipeline stage uses (cheat sheet)

Stage	Forward?	Backward?	Notes
512 OER clean	—	—	Drop null <code>snr_rater</code> ; raw OER rows
Cell 4 merge	Builds both <code>_fwd</code> and <code>_bwd</code> columns	Same	Every snapshot gets both stamps
Cell 5 pool mean/size	<code>tb_ratio_fwd_snr</code> in pool mean	<code>snr_rater_bwd</code> + <code>snpsht_dt</code> for group	Mixed — this is the design choice
Cell 6 pool ranks/z	Same as Cell 5	Same	
Run 1 Cox	<code>z_tb_ratio_fwd_snr</code> , <code>z_pool_minus_mean_snr_fwd</code>	<code>snr_rater_bwd</code> exported for overlap plots only	Model does not include SNR ID as covariate
HERO / BDP panel 6	<code>pool_size_snr_fwd</code>	Overlap uses <code>snr_rater_bwd</code>	Panel 6 histogram = forward size, backward key

7. Recommended fix (opinion — for discussion)

Option A (recommended first step): Re-key pools to rating cycle, not snapshot day

Change in Cell 5 (`add_pool_means_and_sizes` call or function default):

OLD pool key: `(snpsht_dt, snr_rater_bwd)`
NEW pool key: `(snr_rater_bwd, eval_thru_dt_bwd)`

Optional tighten:

`(snr_rater_bwd, eval_strt_dt_bwd, eval_thru_dt_bwd)`

Keep unchanged (for Run 1):

- Own performance: `tb_ratio_fwd_snr` / `z_tb_ratio_fwd_snr`
- Pool comparison values: still average peers’ `tb_ratio_fwd_snr` (forward ratios in the same true rating pool)
- Pool membership ID: still `snr_rater_bwd` (backward SNR — correct for “who is rating this cycle”)

Why this is the right first fix

- Probe already shows p95 drops from **79** → **33** with only `eval_thru_dt_bwd` added.
- Minimal conceptual change: “pool = SNR’s officers in **this** OER cycle,” not “everyone on March 31 with the same SNR person somewhere in their file.”
- Matches Army `snr_rater_rates_this_grd` scale much better.

Simple before/after example

Officer	snpsht_dt	snr_rater_bwd	eval_thru_dt_bwd	OLD poolmates	NEW poolmates
A	2018-03-31	G	2018-09-30	All {A,B,C,...} on 2018-03-31 with G	Officers with G and thru 2018-09-30
D	2018-03-31	G	2017-09-30	Same huge bucket as A	Different, smaller bucket

Option B (config-only experiment): Set `TB_ASSIGNMENT_MODE_POOL = 'backward'`

This switches pool column names to `pool_minus_mean_snr_bwd` etc., averaging `tb_ratio_bwd_snr` within pools. That is a **different scientific question** (“LOO within active-window ratios”), not the same as fixing pool membership. **Do not confuse with Option A.** Option A fixes *who is in the pool*; Option B changes *which performance number is compared*.

Option C (later): Add UIC or rating unit to the pool key

If Option A still leaves tails (max 766 in probe), add `oer_uic_pde` or `snr_rater_asg_uic_pde` from OER. Army pools are often “SNR + unit + cycle.” Defer until Option A is implemented and re-probed.

8. Implementation sketch (when you green-light code)

- `add_cum_oer_metrics_mod_working.py` — allow `group_cols` for SNR pools to include `eval_thru_dt_bwd` (and optionally `eval_strt_dt_bwd`), not `snapshot_date_col`.
 - 520** Cell 5 — pass the new grouping columns; document in cell markdown.
 - Re-run** from Cell 5 → 11 (feather), then BDP/HERO/overlap scripts.
 - Re-run** `army_pool_size_probe.py` — computed median/p95 should move toward official/alt rows.
 - Sanity** — spot-check a few `(snr_rater_bwd, eval_thru_dt_bwd)` groups against raw OER counts for that SNR in that cycle.
- Do not** need to re-run 512 unless OER clean rules change.

9. What you can tell Alex (one paragraph)

“We assign OERs to HRC snapshots in two ways: forward for cumulative performance known at the date, backward for the active rating window. Run 1 models own forward SNR top-block ratio against forward pool-minus-mean, but we keyed pools on snapshot date plus backward SNR ID, which merged officers across rating cycles. Probe shows official OER pool sizes median ~12 vs our computed ~23 with p95 79; re-keying to SNR plus eval-through date drops p95 to ~33. Next step is Cell 5 pool grouping fix and pipeline refresh.”

10. Files to keep handy

File	Role
<code>talent/talent_pipeline/add_cum_oer_metrics_mod_working.py</code>	Forward/backward merge + <code>add_pool_means_and_sizes</code>
<code>talent/talent_pipeline/pipeline_config.py</code>	<code>TB_ASSIGNMENT_MODE_SELF</code> / <code>_POOL</code>
<code>talent/talent_pipeline/520_pipeline_cox_working.ipynb</code>	Cells 4–6
<code>talent/re_entry/army_pool_size_probe.py</code>	Before/after probe
<code>talent/re_entry/army_basic_plots.py</code>	Panel 6 pool size histogram

11. Parked until after fix

- Pool interval overlap panel (`snpsht_dt × snr_rater_bwd`) — update grain to match new pool key.
- HERO / Cox Run 1 — re-estimate after new LOO; coefficients may shift.
- Official `snr_rater_rates_this_grd` — add filter (e.g. 1–200) before validation plots.

Thread log: 2026-09-20 — written after AWS probe; pool-size concern unparked with diagnosis.